# Contents

# 1 2. Homogeneous Differential Equations

When the variables in a first-order differential equation cannot be separated directly (i.e., you cannot get all $x$'s on one side and $y$'s on the other), the next method to check is whether the equation is **Homogeneous**.

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## 1.1 1. Concept Overview

### *1.1.1 1.1 What is a Homogeneous Function?*

A function $F(x, y)$ is called homogeneous of degree $n$ if, when you replace $x$ with $\lambda x$ and $y$ with $\lambda y$, the parameter $\lambda$ can be factored out completely as $\lambda^n$.

$$F(\lambda x, \lambda y) = \lambda^n F(x, y)$$

### *1.1.2 1.2 What is a Homogeneous Differential Equation?*

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is homogeneous if the function $f(x, y)$ can be expressed as a ratio of two homogeneous functions of the **same degree**.

Alternatively, if you can write the differential equation in the form:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

then it is a Homogeneous Differential Equation.

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## 1.2 2. Working Rule (The Substitution Method)

To solve a homogeneous differential equation, we use a specific substitution to transform it into a **Variable Separable** equation.

1. **Identify Homogeneity:** Check that every term in the numerator and denominator has the same total power of variables.
2. **Substitute:**
   - Let $y = vx$
   - Differentiate with respect to $x$ (using the product rule):
   $$\frac{dy}{dx} = v + x\frac{dv}{dx}$$
3. **Transform:** Replace $y$ and $\frac{dy}{dx}$ in the original equation. The $x$ terms will typically cancel out, leaving an equation in terms of $v$ and $x$.
4. **Separate and Integrate:** The new equation will be Variable Separable. Separate $v$ and $x$, then integrate both sides.
5. **Resubstitute:** Finally, replace $v$ with $\frac{y}{x}$ to get the general solution in terms of $x$ and $y$.

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## 1.3 3. Solved Questions

Here are questions from the lecture notes demonstrating the homogeneity check and the solution process.

#### *1.3.1 Question 1: Checking for Homogeneity*

**Determine if the function $F(x,y) = \frac{2x^2y}{x^2+y^2}$ is homogeneous.**

**Solution:**

1. **Test:** Replace $x \to \lambda x$ and $y \to \lambda y$.

$$F(\lambda x, \lambda y) = \frac{2(\lambda x)^2(\lambda y)}{(\lambda x)^2 + (\lambda y)^2}$$

2. **Simplify:**

$$= \frac{2\lambda^2 x^2 \cdot \lambda y}{\lambda^2 x^2 + \lambda^2 y^2}$$
$$= \frac{\lambda^3(2x^2y)}{\lambda^2(x^2+y^2)}$$

3. **Factor:**

$$= \lambda^{3-2}\left(\frac{2x^2y}{x^2+y^2}\right) = \lambda^1 F(x,y)$$

4. **Conclusion:** Yes, it is a homogeneous function of degree **1**.

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#### *1.3.2 Question 2: Solving a Homogeneous DE*

**Solve:** $(2xy + x^2)\frac{dy}{dx} = 3y^2 + 2xy$

**Solution:**

1. **Rewrite in standard form:**

$$\frac{dy}{dx} = \frac{3y^2 + 2xy}{2xy + x^2}$$

*Observation: Every term ($y^2, xy, x^2$) is of degree 2. Thus, it is Homogeneous.*

2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$. Substitute into the equation:

$$v + x\frac{dv}{dx} = \frac{3(vx)^2 + 2x(vx)}{2x(vx) + x^2}$$

3. **Simplify the Fraction:**

$$v + x\frac{dv}{dx} = \frac{x^2(3v^2 + 2v)}{x^2(2v+1)}$$

$$v + x\frac{dv}{dx} = \frac{3v^2 + 2v}{2v+1}$$

4. **Separate Variables:** Move $v$ to the right side:

$$x\frac{dv}{dx} = \frac{3v^2 + 2v}{2v+1} - v$$

Find a common denominator:

$$x\frac{dv}{dx} = \frac{3v^2 + 2v - v(2v+1)}{2v+1}$$

$$x\frac{dv}{dx} = \frac{3v^2 + 2v - 2v^2 - v}{2v+1} = \frac{v^2 + v}{2v+1}$$

Now, separate $v$ and $x$:

$$\frac{2v+1}{v^2+v}dv = \frac{1}{x}dx$$

5. **Integrate:**

$$\int \frac{2v+1}{v^2+v}dv = \int \frac{1}{x}dx$$

*Notice: The numerator $(2v+1)$ is the exact derivative of the denominator $(v^2+v)$.*

$$\ln|v^2 + v| = \ln|x| + \ln C$$

$$v^2 + v = Cx$$

6. **Resubstitute $v = y/x$:**

$$\left(\frac{y}{x}\right)^2 + \frac{y}{x} = Cx$$

$$\frac{y^2 + xy}{x^2} = Cx \implies y^2 + xy = Cx^3$$

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*1.3.3 Question 3: Trigonometric Homogeneous DE*

**Solve:** $x\sin\left(\frac{y}{x}\right)dy = \left[y\sin\left(\frac{y}{x}\right) - x\right]dx$

**Solution:**

1. **Rewrite in standard form:**

$$\frac{dy}{dx} = \frac{y\sin(y/x) - x}{x\sin(y/x)}$$

$$\frac{dy}{dx} = \frac{y}{x} - \frac{x}{x\sin(y/x)} = \frac{y}{x} - \frac{1}{\sin(y/x)}$$

*Observation: The equation is a function of $(y/x)$, so it is Homogeneous.*

2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$.

$$v + x\frac{dv}{dx} = v - \frac{1}{\sin v}$$

3. **Simplify:** Subtract $v$ from both sides:

$$x\frac{dv}{dx} = -\frac{1}{\sin v} = -\csc v$$

4. **Separate Variables:**

$$\sin v\, dv = -\frac{1}{x}dx$$

5. **Integrate:**
$$\int \sin v \, dv = -\int \frac{1}{x} dx$$
$$-\cos v = -\ln|x| + C$$
Multiply by $-1$:
$$\cos v = \ln|x| - C$$
6. **Resubstitute $v = y/x$:**
$$\cos\left(\frac{y}{x}\right) = \ln|x| + C'$$

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**Next Method:** What if the equation looks homogeneous but has constant terms interfering? Proceed to Method 3: Equations Reducible to Homogeneous Form.